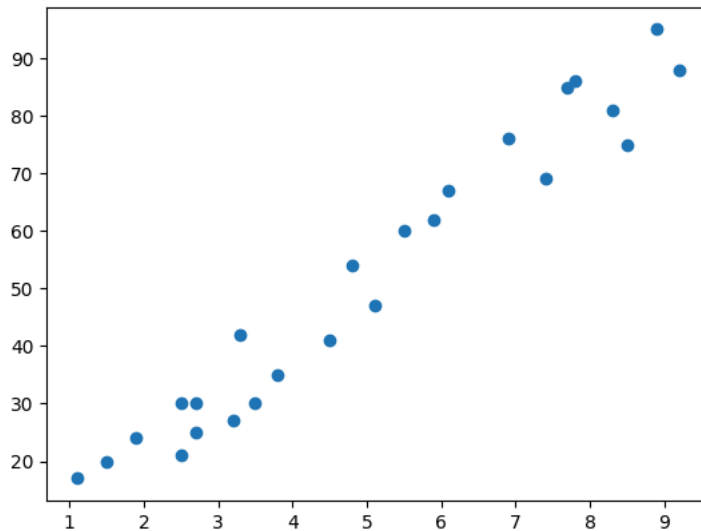


```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
dataset = pd.read_csv('Student_scores.csv')
plt.scatter(dataset['Hours'], dataset['Scores'])
plt.show()
```



```
X = dataset.iloc[:, :-1].values
Y = dataset.iloc[:, 1].values
```

```
print(Y)
```

```
[21 47 27 75 30 20 88 60 81 25 85 62 41 42 17 95 30 24 67 69 30 54 35 76
 86]
```

```
class Model():
    def __init__(self, learning_rate, iterations):
        self.learning_rate = learning_rate
        self.iterations = iterations

    def predict(self, X):
        return X.dot(self.slope) + self.const

    def fit(self, X, Y):
        self.m, self.n = X.shape
        self.slope = np.zeros(self.n)
        self.const = 0
        self.X = X
        self.Y = Y

        for i in range(self.iterations):
            self.update_weights()
        return self

    def update_weights(self):
        Y_pred = self.predict(self.X)
        dw = -(2 * (self.X.T).dot(self.Y - Y_pred)) / self.m
        db = -2 * np.sum(self.Y - Y_pred) / self.m

        self.slope = self.slope - self.learning_rate * dw
        self.const = self.const - self.learning_rate * db
        return self
```

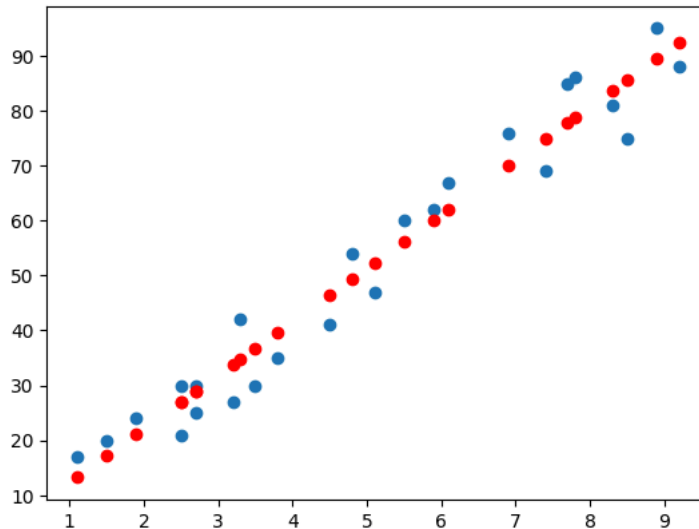
```
model = Model(learning_rate = 0.01, iterations = 1000)
model.fit(X, Y)
```

```
<__main__.Model at 0x7d4c0afda360>
```

```
Y_pred = model.predict(X)
print(Y_pred)
```

```
[26.91171724 52.33687281 33.75695143 85.58515317 36.69062323 17.13281125
92.43038736 56.24843521 83.62937197 28.86749844 77.76202838 60.1599976
46.46952922 34.73484203 13.22124886 89.49671557 26.91171724 21.04437365
62.1157788 74.82835658 28.86749844 49.40320102 39.62429503 69.93890359
78.73991898]
```

```
plt.scatter(dataset['Hours'], dataset['Scores'])
plt.scatter(X, Y_pred, color = 'red')
plt.show()
print(model.slope)
print(model.const)
```



```
[9.77890599]
2.4644522714760995
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

datasets = pd.read_csv('Student_scores.csv')
X = datasets.iloc[:, :-1].values
Y = datasets.iloc[:, 1].values
```

```
print(X)
```

```
[[2.5]
 [5.1]
 [3.2]
 [8.5]
 [3.5]
 [1.5]
 [9.2]
 [5.5]
 [8.3]
 [2.7]
 [7.7]
 [5.9]
 [4.5]
 [3.3]
 [1.1]
 [8.9]
 [2.5]
 [1.9]
 [6.1]
 [7.4]
 [2.7]
 [4.8]
 [3.8]
 [6.9]
 [7.8]]
```

```
from sklearn.linear_model import LinearRegression
regressor = LinearRegression()
```

```
from sklearn.model_selection import train_test_split
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size = 0.2, random_state = 0)
```

```
regressor.fit(X_train, Y_train)
```

LinearRegression ⓘ ?

LinearRegression()

```
y_pred = regressor.predict(X_test)
print(y_pred)
```

```
[16.88414476 33.73226078 75.357018 26.79480124 60.49103328]
```

```
for i, (pred, actual) in enumerate(zip(y_pred, Y_test)):
    print(f"Sample {i+1}: Predicted = {pred:.2f}, Actual = {actual:.2f}")
```

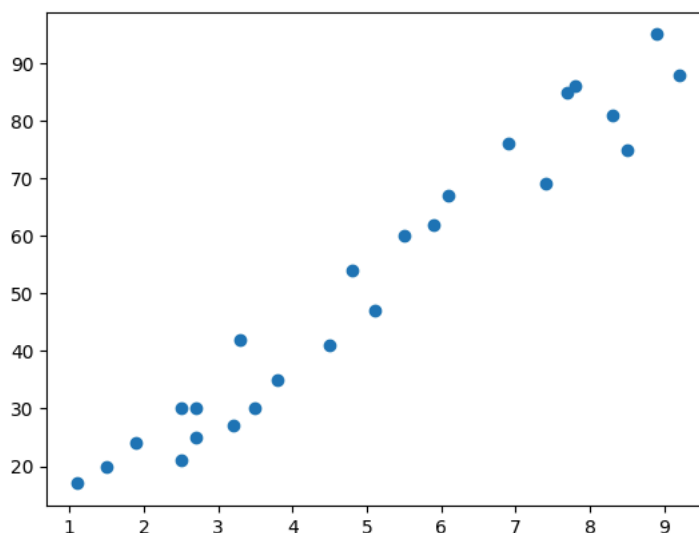
```
Sample 1: Predicted = 16.88, Actual = 20.00
Sample 2: Predicted = 33.73, Actual = 27.00
Sample 3: Predicted = 75.36, Actual = 69.00
Sample 4: Predicted = 26.79, Actual = 30.00
Sample 5: Predicted = 60.49, Actual = 62.00
```

```
print("Coefficients: ", regressor.coef_)
print("Intercept:", regressor.intercept_)
```

```
Coefficients: [9.91065648]
Intercept: 2.018160041434662
```

$$Y = 2.36992324e-16x_1 + 1.00000000e+00x_2 + -2.22044605e-16x_3 + -5.820766091346741e-11$$

```
dataset = pd.read_csv('Student_scores.csv')
plt.scatter(dataset['Hours'], dataset['Scores'])
plt.show()
X = dataset.iloc[:, :-1].values
Y = dataset.iloc[:, -1].values
```



```
from sklearn.model_selection import train_test_split
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size = 0.2, random_state = 0)
```

```
from sklearn.preprocessing import PolynomialFeatures
poly = PolynomialFeatures(degree=3)
X_poly = poly.fit_transform(X)
print(X_poly)
```

```
[[ 1.    2.5    6.25  15.625]
 [ 1.    5.1   26.01  132.651]
 [ 1.    3.2   10.24   32.768]
 [ 1.    8.5   72.25  614.125]
 [ 1.    3.5   12.25   42.875]
 [ 1.    1.5    2.25    3.375]
 [ 1.    9.2   84.64  778.688]
 [ 1.    5.5   30.25  166.375]
 [ 1.    8.3   68.89  571.787]
 [ 1.    2.7    7.29   19.683]
 [ 1.    7.7   59.29  456.533]
 [ 1.    5.9   34.81  205.379]
 [ 1.    4.5   20.25   91.125]
 [ 1.    3.3   10.89   35.937]
 [ 1.    1.1    1.21    1.331]
 [ 1.    8.9   79.21  704.969]
 [ 1.    2.5    6.25   15.625]
```

```
[ 1.    1.9    3.61   6.859]
[ 1.    6.1   37.21  226.981]
[ 1.    7.4   54.76  405.224]
[ 1.    2.7    7.29  19.683]
[ 1.    4.8   23.04  110.592]
[ 1.    3.8   14.44   54.872]
[ 1.    6.9   47.61  328.509]
[ 1.    7.8   60.84  474.552]]
```

```
model = LinearRegression()
model.fit(X_poly, Y)
```

▼ LinearRegression ⓘ ?

```
LinearRegression()
```

```
print(model.coef_)
print(model.intercept_)
```

```
[ 0.          -3.79705913   2.99874775  -0.19257785]
19.317706990502344
```

```
y_pred = model.predict(X_poly)
print(y_pred)
```

```
[25.55820366 52.4044897 31.56390366 85.43535572 34.50588455 19.71935048
 88.24070983 57.10586101 84.27233622 27.13600857 79.9579611 61.75002055
 45.40692604 32.52310457 18.5131056 87.29327399 25.55820366 21.60788253
 64.02753655 77.39372855 27.13600857 48.88540147 37.62366789 72.62482165
 80.75625387]
```

```
plt.scatter(X, Y, color='blue')
plt.scatter(X, y_pred, color='red')
plt.title('Polynomial Regression')
```

```
plt.show()
```

